

## ◆ SECTION 1: BASIC CONCEPTS & TYPES OF SOLUTIONS (Q1–Q25)

### 1. Correct: B – Homogeneous mixture

A solution has **uniform composition throughout**, unlike heterogeneous mixtures.

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### 2. Correct: A – One solute and one solvent only

Binary = **two components only** (solute + solvent).

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### 3. Correct: B – Solute

Sugar is present in **smaller amount**, so it is solute.

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### 4. Correct: C – Solid in solid solution

Brass is an alloy → **solid solute (Zn) in solid solvent (Cu)**.

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### 5. Correct: A – Gas in gas solution

Air = mixture of gases ( $\text{N}_2$  solvent,  $\text{O}_2$  etc. solutes).

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### 6. Correct: A – Alcohol in water

Both are liquids → **liquid–liquid solution**.

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### 7. Correct: B – Gas in solid solution

Hydrogen gas dissolved in solid palladium.

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### 8. Correct: B

Saturated solution contains **maximum solute in equilibrium with undissolved solute** at that temperature.

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### 9. Correct: C

Unsaturated solution contains **less solute than solubility limit**.

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**10. Correct: A – Always unstable**

Supersaturated solutions are **metastable** and crystallize easily.

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**11. Correct: B – Sugar in water**

Aqueous solution → **water as solvent**.

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**12. Correct: C – Iodine in ethanol**

Solvent is **not water**, hence non-aqueous.

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**13. Correct: A – In smaller amount by moles**

Definition of solute is based on **amount**, not state.

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**14. Correct: A**

Solvent determines **physical state** of solution.

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**15. Correct: B – Homogeneous**

Miscible liquids mix uniformly.

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**16. Correct: B – 2**

Binary means **two components**.

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**17. Correct: B – Gas in liquid**

CO<sub>2</sub> dissolved in water.

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**18. Correct: D – Iodine in water – liquid in liquid (wrong)**

Iodine is **solid**, so correct type is solid in liquid.

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**19. Correct: C – True solution**

True solutions are **always homogeneous**.

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**20. Correct: B – Size of solute particles**

Main difference between colloids and true solutions.

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**21. Correct: B**

Solute = component present in **smaller amount**.

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**22. Correct: B – Less volatile component**

When amounts are equal, **less volatile = solvent** (NCERT rule).

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**23. Correct: B – Sand suspended in water**

Suspension  $\neq$  solution.

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**24. Correct: A – Binary solutions**

Exactly **one solute + one solvent**.

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**25. Correct: A – Filtration**

True solutions **pass through filter paper**.

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◆ **SECTION 2: CONCENTRATION UNITS (Q26–Q60)**

**26. Correct: B**

$$\text{Mass \%} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

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**27. Correct: A**

$$X_A = \frac{n_A}{n_A + n_B}$$

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28. Correct: B

$$M = \frac{\text{moles of solute}}{\text{litre of solution}}$$

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29. Correct: B

$$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$$

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30. Correct: B – mol L<sup>-1</sup>

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31. Correct: A – mol kg<sup>-1</sup>

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32. Correct: C – Mole fraction

It is dimensionless.

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33. Correct: B – Molarity

Volume changes with temperature.

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34. Correct: B – Molality

Independent of temperature → best for colligative properties.

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35. Correct: B – Molality

Uses mass, not volume.

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36. Correct: A

$$n = M \times V$$

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37. Correct: B

1 mol solute in 1 kg solvent.

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**38. Correct: C**

1 mol solute in 1 L of solution.

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**39. Correct: B**

On heating, volume  $\uparrow \rightarrow$  molarity  $\downarrow$ .

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**40. Correct: B**

0.9 g per 100 mL solution.

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**41. Correct: A**

For dilute aqueous solutions,  $M \approx m$ .

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**42. Correct: B**

ppm used for **very dilute solutions** (pollutants).

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**43. Correct: B**

10 g NaCl per 100 g solution.

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**44. Correct: C**

$$X_{\text{solvent}} = 1 - X_{\text{solute}} = 0.8$$

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**45. Correct: B**

Sum of mole fractions = 1 always.

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**46. Correct: C – Molality**

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**47. Correct: B – Normality**

Based on **equivalents**, convenient for titrations.

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**48. Correct: D – ppm**

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**49. Correct: B**

0.5 mol per L of solution.

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**50. Correct: C – Molarity**

Osmotic pressure:

$$\pi = CRT$$

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**51. Correct: C – Atom percentage**

Not used in solution chemistry.

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**52. Correct: B**

Always:

$$X_{solute} + X_{solvent} = 1$$

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**53. Correct: A**

$$M_1 V_1 = M_2 V_2 \Rightarrow V_1 = \frac{0.1 \times 250}{1} = 25 \text{ mL}$$

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**54. Correct: B – Molality**

Density not required.

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**55. Correct: B – Molarity**

Directly relates moles with volume.

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56. Correct: C

$\text{g L}^{-1}$  = mass per litre of solution.

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57. Correct: A

$$m = \frac{0.02}{0.5} = 0.04\text{ } m$$

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58. Correct: D

$$M = \frac{0.5}{0.25} = 2.0\text{ } M$$

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59. Correct: C – Molality

Independent of pressure & temperature.

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60. Correct: A – Mole percent

$$\text{Mole \%} = X \times 100$$

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### ◆ SECTION 3: SOLUBILITY & HENRY'S LAW (Q61–Q90)

61. Correct: A

Definition of solubility at given temperature.

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62. Correct: B

Most solids dissolve more at **higher temperature**.

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63. Correct: B

Gas solubility ↓ when temperature ↑.

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64. Correct: B

Henry's law:

$$p \propto \text{solubility}$$

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**65. Correct: C**

$$p = k_H c$$

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**66. Correct: B**

p = partial pressure of gas.

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**67. Correct: B**

c = concentration of dissolved gas.

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**68. Correct: B**

Large  $k_H \rightarrow$  low solubility.

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**69. Correct: A**

Direct proportionality.

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**70. Correct: C – O<sub>2</sub>**

Least reactive  $\rightarrow$  best obeys Henry's law.

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**71. Correct: B**

Valid for **slightly soluble gases** at low pressure.

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**72. Correct: D**

Depends on form used.

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**73. Correct: B**



High pressure + low temperature  $\uparrow$  solubility.

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**74. Correct: B**

Excess  $\text{N}_2$  dissolves in blood.

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**75. Correct: C**

Pressure  $\uparrow \rightarrow$  solubility  $\uparrow$ .

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**76. Correct: B**

Cold water holds more  $\text{CO}_2$ .

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**77. Correct: A**

Common lab units.

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**78. Correct: D**

Colour irrelevant.

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**79. Correct: B**

$\text{O}_2$  solubility governed by Henry's law.

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**80. Correct: A**

Pressure drops  $\rightarrow$  gas escapes.

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**81. Correct: B**

Only partial pressure matters.

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**82. Correct: A**

Raoult's law = Henry's law special case.

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**83. Correct: C**

Fails at high pressure or non-ideal gases.

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**84. Correct: A**

Straight line through origin.

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**85. Correct: B**

Higher T → gas less soluble →  $k_H \uparrow$ .

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**86. Correct: C**

Depends on enthalpy of dissolution.

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#### ◆ SECTION 4: RAOULT'S LAW & IDEAL SOLUTIONS (Q91–Q115)

**91. Correct: B**

$$p = x_{\text{solvent}}p^\circ$$

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**92. Correct: A**

Raoult's law for volatile component.

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**93. Correct: A**

Dalton's law applies.

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**94. Correct: A**

Ideal solutions obey law at all compositions.

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**95. Correct: C**

No heat absorbed or evolved.

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**96. Correct: A**

No volume change on mixing.

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**97. Correct: B**

Weaker A–B attractions → molecules escape easily.

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**98. Correct: B**

Stronger A–B attractions → vapour pressure ↓.

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**99. Correct: A**

Higher vapour pressure = positive deviation.

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**100. Correct: B**

Lower vapour pressure = negative deviation.

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**101. Correct: C**

Chemically similar liquids.

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**102. Correct: B**

H-bond breaking → positive deviation.

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**103. Correct: C**

Strong H-bonding.

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**104. Correct: B**

Higher vapour pressure → lower boiling point.

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**105. Correct: B**

Lower vapour pressure → higher boiling point.

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**106. Correct: B**

Boil at constant temperature.

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**107. Correct: A**

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**108. Correct: B**

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**109. Correct: B**

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**110. Correct: C**

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**111. Correct: A**

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**112. Correct: A**

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**113. Correct: B**

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**114. Correct: C**

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**115. Correct: D**

Strong chemical interaction  $\neq$  ideal.

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◆ **SECTION 5: COLLIGATIVE PROPERTIES (Q116–Q150)**

**116. Correct: B**

Depends only on **number of particles**.

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117. Correct: D

Viscosity is **not** colligative.

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118. Correct: A

$$\frac{p^{\circ} - p}{p^{\circ}}$$

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119. Correct: B

$$\Delta T_b = K_b m$$

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120. Correct: A

$$\Delta T_f = K_f m$$

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121. Correct: A

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122. Correct: C

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123. Correct: B

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124. Correct: B

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125. Correct: B

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126. Correct: A

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127. Correct: C

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128. Correct: C

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129. Correct: B

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130. Correct: B

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131. Correct: A

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132. Correct: A

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133. Correct: B

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134. Correct: A

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135. Correct: B

More particles → dissociation.

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136. Correct: A

Fewer particles → association.

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137. Correct: D

Most sensitive for dilute solutions.

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138. Correct: B

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139. Correct: C

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140. Correct: B

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141. Correct: B

NaCl  $\rightarrow$  i  $\approx$  2.

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142. Correct: C

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143. Correct: A

For AB<sub>2</sub>:

$$i = 1 + 2\alpha \Rightarrow \alpha = \frac{i - 1}{2}$$

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144. Correct: B

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145. Correct: A

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146. Correct: B

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147. Correct: B

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148. Correct: C

$$\pi \propto C$$

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149. Correct: C

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150. Correct: A

Colligative properties depend on **number of moles of solute**.